# COURSEWORK ASSIGNMENT SUBMISSION

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**Subject:** Software Engineering & Project Management **Day:** Day 18 — Software Testing Fundamentals

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## Section 10: Classroom Activity — Test CEMS

Design and execution logging of structured test cases for the CEMS event registration engine covering successful paths, edge cases, and non-functional validation.

| Test ID | Category | Preconditions | Execution Steps & Inputs | Expected Output | Pass/Fail |
|---|---|---|---|---|---|
| TC-REG-01 | Happy Path | Event #101 active, capacity 50/100. | Student A clicks "Register", submits valid credentials. | HTTP 200, confirmation code generated, seat count becomes 51. | PASS |
| TC-REG-02 | Happy Path | Event #105 has open seats, no conflicts. | Student registers for 2nd non-overlapping event. | HTTP 200, registration logged successfully. | PASS |
| TC-REG-03 | Edge Case | Event #102 capacity is exactly 100/100. | Student attempts registration on full event. | HTTP 409 Conflict, message "Registration full", count remains 100. | PASS |
| TC-REG-04 | Edge Case | Student already registered for Event #101. | Student attempts duplicate registration on Event #101. | HTTP 400 Bad Request, message "Already registered". | PASS |
| TC-REG-05 | Edge Case | Student registered for Event at 2-4 PM. | Attempts registration for Event at 3-5 PM. | Validation blocks submission, displays schedule conflict alert. | PASS |
| TC-NFR-01 | Non-Functional | 200 concurrent virtual users active. | Trigger registration API flood over 60 seconds. | p95 latency < 1.5s, 0% database deadlock errors. | PASS |

## Section 13: Homework — Testing Fundamentals Packet

Testing theory, error taxonomy, and formal defect reporting for library automation.

## 1. Error vs Defect vs Failure Taxonomy

| Scenario | Human Error (Mistake) | Internal Defect (Fault / Bug) | Observed System Failure |
|---|---|---|---|
| Boundary Comparison | Developer typed `<` instead of `<=` in capacity validation. | Source code logic checks `if (count < max)` omitting the final seat. | System rejects the 100th attendee when maximum capacity is 100. |
| Null Pointer in Email | Developer forgot to verify if student has an email registered. | Missing null check prior to invoking `smtpClient.send()`. | Server throws HTTP 500 Internal Error during checkout confirmation. |
| Un-indexed Foreign Key | Database architect omitted indexing on `student_id` column. | Full table scan executed on every registration lookup query. | Portal response time degrades to 12 seconds during morning traffic rush. |

## 2. Formal Defect Report Specification

- **Defect ID:** BUG-CEMS-2026-042
- **Summary:** Simultaneous last-seat registration allows 101 attendees in 100-capacity workshop.
- **Severity / Priority:** Severity: Critical (Data Integrity) / Priority: High.
- **Environment:** Staging Cluster (Node.js v20, PostgreSQL 16, Ubuntu 24.04).
- **Reproduction Steps:** 1. Set event #402 capacity to 100. 2. Pre-fill 99 registrations. 3. Fire two concurrent registration requests at the identical millisecond timestamp. 4. Query `SELECT COUNT(*) FROM registrations WHERE event_id = 402`.
- **Observed Result:** Both requests return HTTP 200 OK; database record count is 101.
- **Expected Result:** One request succeeds (200 OK); the second request returns HTTP 409 Conflict; count remains 100.
- **Root Cause:** Missing database row-level lock (`FOR UPDATE`) in capacity verification transaction.

DRAFT